

## Ejercicio 1.a: Demostrar que State es una Mónica.

\* Para realizar dicha demostración, debemos probar que cumplen las 3 leyes de mónicas.

### Monad.1:

$$\begin{aligned} & \bullet \text{return } a \gg= \kappa \quad \langle \text{def. return} \rangle \\ = & \bullet \text{State } (\lambda s \rightarrow (a :: s)) \gg= \kappa \\ = & \bullet \text{State } (\lambda s \rightarrow \text{let } (m :: s') = \text{runState } (\text{State } (\lambda s'' \rightarrow (a :: s''))) s \\ & \quad \text{in runState } (\kappa m) s') \quad \langle \text{def. } \gg= \rangle \\ = & \bullet \text{State } (\lambda s \rightarrow \text{let } (m :: s') = (\lambda s'' \rightarrow (a :: s'')) s \\ & \quad \text{in runState } (\kappa m) s') \quad \langle \text{runState.State} = \text{id} \rangle \\ = & \bullet \text{State } (\lambda s \rightarrow \text{let } (m :: s') = (a :: s) \\ & \quad \text{in runState } (\kappa m) s') \quad \langle \text{def. App} \rangle \\ = & \bullet \text{State } (\lambda s \rightarrow \text{runState } (\kappa a) s) \quad \langle \text{def. Let} \rangle \\ = & \bullet \text{State } (\text{runState } (\kappa a)) \quad \langle \eta\text{-red} \rangle \\ = & \bullet (\kappa a) \quad \langle \text{runState.State} = \text{id} \rangle \end{aligned}$$

Wego,  $\text{return } a \gg= \kappa = \kappa a$ . Por lo que vale Monad.1

## Monad.2.

•  $m \gg = \text{return}$

$\langle \text{def. } \gg = \rangle$

= •  $\text{State } (\lambda s \rightarrow \text{let } (m!:s') = \text{runState } m \ s$   
      $\text{in } \text{runState } (\text{return } m) \ s')$

$\langle \text{def. return} \rangle$

= •  $\text{State } (\lambda s \rightarrow \text{let } (m!:s') = \text{runState } m \ s$   
      $\text{in } \text{runState } (\text{State } (\lambda s'' \rightarrow (x!:s'')))) \ s')$

$\langle \text{runState, State} = \text{id} \rangle$

= •  $\text{State } (\lambda s \rightarrow \text{let } (m!:s') = \text{runState } m \ s$   
      $\text{in } (\lambda s'' \rightarrow (m!:s'')) \ s')$

$\langle \text{def. App} \rangle$

= •  $\text{State } (\lambda s \rightarrow \text{let } (m!:s') = \text{runState } m \ s$   
      $\text{in } (m!:s')$

$\langle \text{def. let} \rangle$

= •  $\text{State } (\lambda s \rightarrow \text{runState } m \ s)$

$\langle \eta\text{-red} \rangle$

= •  $\text{State } (\text{runState } m)$

$\langle \text{State, runState} = \text{id} \rangle$

= •  $m$

Wego obtuvimos que:  $m \gg = \text{return} = m$

∴ Vale Monad.2



# Monad. 3

Separaremos la prueba, desarrollando ambos ramos de la igualdad, para llegar al mismo termino y luego aplicar transitividad.

$$A). (m \gg= k) \gg= h$$

$\langle \text{def. } \gg= \rangle$

$$= \cdot \text{State } (\lambda s1 \rightarrow \text{let } (m :! : s1') = \text{runState } m \ s1 \\ \text{in } \text{runState } (k \ m) \ s1') \gg= h$$

$\langle \text{def. } \gg= \rangle$

$$= \cdot \text{State } (\lambda s \rightarrow \text{let } (m :! : s') = \text{runState } (\text{State } (\lambda s1 \rightarrow \text{let } (m1 :! : s1') = \text{runState } m \ s1 \\ \text{in } \text{runState } (k \ m1) \ s1')) \ s) \\ \text{in } \text{runState } (h \ m) \ s')$$

$\langle \text{runState. State} = \text{id} \rangle$

$$= \cdot \text{State } (\lambda s \rightarrow \text{let } (m :! : s') = (\lambda s1 \rightarrow \text{let } (m1 :! : s1') = \text{runState } m \ s1 \\ \text{in } \text{runState } (k \ m1) \ s1') \ s) \\ \text{in } \text{runState } (h \ m) \ s')$$

$\langle \text{def. App} \rangle$

$$= \cdot \text{State } (\lambda s \rightarrow \text{let } (m :! : s') = (\text{let } (m1 :! : s1') = \text{runState } m \ s \\ \text{in } \text{runState } (k \ m1) \ s1') \\ \text{in } \text{runState } (h \ m) \ s')$$

$\langle \text{Prop. Let} \rangle$

$$= \cdot \text{State } (\lambda s \rightarrow \text{let } (m1 :! : s1') = \text{runState } m \ s. \\ (m :! : s') = \text{runState } (k \ m1) \ s1' \\ \text{in } \text{runState } (h \ m) \ s') \quad \left. \vphantom{\text{State}} \right\} \textcircled{1}$$

B)

$$= \cdot m \gg = (\lambda x \rightarrow \kappa x \gg = h)$$

<def. >>=>

$$= \cdot \text{State} (\lambda s \rightarrow \text{let } (m1 :! : s1') = \text{runState } m \ s \cdot \\ \text{in } \text{runState} ((\lambda x \rightarrow \kappa x \gg = h) m) s')$$

<def. App>

$$= \cdot \text{State} (\lambda s \rightarrow \text{let } (m1 :! : s1') = \text{runState } m \ s \\ \text{in } \text{runState} (\kappa m \gg = h) s')$$

<def. >>=>

$$= \cdot \text{State} (\lambda s \rightarrow \text{let } (m1 :! : s1') = \text{runState } m \ s \\ \text{in } \text{runState} (\text{State} (\lambda s'' \rightarrow \text{let } (m :! : s') = \text{runState} (\kappa m) s'' \\ \text{in } \text{runState} (h m) s') \\ ) s')$$

<runState.State = id>

$$= \cdot \text{State} (\lambda s \rightarrow \text{let } (m1 :! : s1') = \text{runState } m \ s \\ \text{in } (\lambda s'' \rightarrow \text{let } (m :! : s') = \text{runState} (\kappa m) s'' \\ \text{in } \text{runState} (h m) s') \\ s')$$

<def. App>

$$= \cdot \text{State} (\lambda s \rightarrow \text{let } (m1 :! : s1') = \text{runState } m \ s \cdot \\ \text{in } \text{let } (m :! : s') = \text{runState} (\kappa m) s1' \\ \text{in } \text{runState} (h m) s')$$

<def. Let>

$$= \cdot \text{State} (\lambda s \rightarrow \text{let } (m1 :! : s1') = \text{runState } m \ s \\ (m :! : s') = \text{runState} (\kappa m) s1' \\ \text{in } \text{runState} (h m) s') \quad \left. \vphantom{\text{State}} \right\} \textcircled{2}$$

wego por ① y ②, Monad.3 vale!!

∴ State es efectivamente una Monada